SWAPNEEL PUROHIT

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SUMMARY

Infrastructure engineer with hands-on production experience in Linux server administration, containerized deployments, and AI server infrastructure. Currently manage Cloudflare Tunnel configurations, Docker deployments, SSH-based remote administration, and monitoring stack rollout (Prometheus, Grafana) at BigWings LLC, alongside AWS-to-private-infrastructure migration and security hardening for production systems. Backend development background in Node.js, Django, and FastAPI handling authentication, real-time communication, and structured data pipelines across four shipped platforms. Comfortable operating across the stack from server provisioning and network configuration to API design and database architecture.

EDUCATION

B.Tech - Computer Science and System Engineering | CGPA: 8.4

Kalinga Institute of Industrial Technology (KIIT), Bhubaneswar | 2023 – 2027

Class XII - CBSE (PCMB) | Percentage: 84.8

Kendriya Vidyalaya, Cuttack | 2021-2023

Class X - CBSE | Percentage: 95.3

Kendriya Vidyalaya, Angul | 2021

TECHNICAL SKILLS

Cloud & DevOps:EC2, RDS, S3, CloudFront, VPC, IAM, Route 53, CloudWatch; working knowledge of additional AWS services, AWS-to-Private Infra Migration, Docker, Podman, Cloudflare Tunnel, CI/CD, GitHub Actions, Prometheus, Grafana,Node Exporter, Nginx.
Infrastructure & Systems: Linux Server Administration, SSH Remote Administration, NoMachine, Network Configuration, Firewall Configuration (UFW), DNS Configuration, Systemd, Security Hardening, Infrastructure Monitoring.
Languages: Java, Python, Javascript, SQL, C
Backend & APIs: Node.js, Express.js, Django, Flask, REST APIs, WebSockets, JWT Authentication
Databases: PostgreSQL, MySQL, Redis, Firebase
Frontend: React.js, Next.js, HTML5, CSS3, TailwindCSS
Software Engineering: System Design, OOP, Data Structures & Algorithms, Scalable Architecture, SaaS Applications
AI/ML: TensorFlow, scikit-learn, NumPy, Pandas, OpenAI API, LLM Integration, Hugging Face, NLP, Ollama, BERT.
Tools: Git/GitHub, Postman, Celery, ngrok, Agile Development

EXPERIENCE

AI Infrastructure & Cloud Intern | BigWings LLC

May *2026 – Present*

- **Audit and stabilize AWS infrastructure**, reviewing existing cloud resources and configurations, documenting the environment, identifying operational gaps, and implementing corrective changes across production services.
- **Execute AWS-to-on-premises infrastructure migration**, transitioning production workloads and services including **EC2 and RDS** to company-controlled infrastructure while contributing to the design, provisioning, and operational readiness of the private environment.
- **Provision and administer Linux-based AI infrastructure**, configuring **headless servers**, managing systems remotely through **SSH**, troubleshooting services and connectivity, and maintaining production readiness across server environments.
- **Configure NVIDIA GPU servers and server-rack infrastructure** for AI workloads, covering hardware configuration, system onboarding, inter-server networking, deployment preparation, and hardware/software readiness.
- **Implement infrastructure security and network controls** through **SSH hardening, firewall configuration, secure routing, IP/network management, and Cloudflare Tunnel**, enabling controlled remote access while minimizing unnecessary public exposure.
- **Build centralized infrastructure observability** for the on-premises environment using **Prometheus, Grafana, Node Exporter, and cAdvisor**, establishing visibility across compute, memory, storage, network, and container resources and troubleshooting monitoring-stack reliability.
- **Deploy and maintain containerized infrastructure with Docker**, configuring service networking, persistent storage, runtime dependencies, and monitoring integrations while documenting repeatable deployment and operational procedures.

Backend Developer & System Integration Contributor | Bliski (Client Project)
October 2025 – February 2026
github link: *https://github.com/easyqr-org/EasyQR*

Engineered and delivered a production-grade **QR-based session pairing + secure redirection system (EasyQR)** enabling seamless desktop–mobile device linking and dynamic URL routing workflows.
Designed and implemented scalable **RESTful backend services** for QR generation, decoding, session validation, and secure redirection, with **strict input validation**, structured exception handling, and standardized API responses.
Built a robust **session lifecycle engine** (PENDING → CONNECTED → TERMINATED) with controlled expiry and state enforcement, ensuring secure pairing flows and preventing unauthorized reuse of sessions.
Integrated **bidirectional WebSocket communication** for real-time low-latency event delivery, enabling stable device-to-device messaging and session synchronization.
Collaborated in iterative delivery cycles involving debugging, regression testing, and integration hardening to ensure **deployment-ready stability** and alignment with real-world operational requirements.

Co-Founder & Full-Stack Developer | Vartaman.tech (*2026 – Present)*
Link: *https://www.vartaman.tech/*

- Co-founded a technology startup delivering **10+ scalable business websites and digital platforms** for startups, professionals, and local businesses.
- Built responsive full-stack applications using **React.js, Next.js, Django, Node.js, PostgreSQL, REST APIs, and AWS**.
- Improved client digital presence through **SEO-focused, performance-optimized, and mobile-responsive solutions**, contributing to increased lead generation and business visibility.
- Managed deployment workflows including **domain setup, DNS configuration, cloud hosting, authentication systems, and production maintenance**.

PROJECTS

InternPilot — AI-Powered Internship Management Platform
GitHub: https://github.com/BlackPrince2607/InternPilot | Live Demo: https://internpilot-frontend.vercel.app
React.js, FastAPI, Node.js, PostgreSQL, Supabase, Groq, TailwindCSS, REST APIs

- Developed a production-style full-stack internship management platform with **JWT-authenticated workflows**, supporting **resume uploads, semantic job matching, application tracking, and AI-powered cold email generation**.
- Engineered scalable backend systems using **FastAPI, PostgreSQL, Supabase Auth & Storage**, and **REST APIs**, implementing structured resume parsing pipelines with **pdfplumber**, **Groq LLM integration**, and JSON-based data extraction.
- Built intelligent job recommendation workflows using **sentence-transformers embeddings**, multi-source job ingestion pipelines, and preference-aware ranking systems for optimized internship matching.
- Designed and managed **9+ normalized database entities** including users, resumes, jobs, applications, outreach tracking, and analytics systems with modular backend architecture and admin-controlled ingestion pipelines.
- Developed responsive frontend dashboards using **React.js, TailwindCSS, and Axios**, enabling centralized tracking of applications, outreach activity, resume analytics, and workflow productivity.

VeriSight — AI Deepfake Detection Platform
GitHub: https://github.com/DEVELOPER-SWAPNEEL/Verisight | Live Demo:**https://verisight-production.up.railway.app**
Python, FastAPI, PyTorch, Redis, MySQL, OpenCV, WebSockets, Docker

- Built a distributed AI-powered deepfake detection platform supporting **image, video, and live-stream analysis** using **FastAPI microservices, Redis job queues, and PyTorch/CUDA inference pipelines**.
- Engineered scalable asynchronous processing workflows with **WebSocket-based real-time updates**, background workers, and confidence-based result aggregation for low-latency detection.
- Designed and managed a production-style backend architecture integrating **MySQL, SQLAlchemy, OpenCV, Docker, and multi-service communication** for secure and maintainable system operations.

CERTIFICATIONS

- AWS Academy Graduate Cloud Foundations & Microservices and CI/CD Pipeline Builder – AWS
- Cisco Certified Ethical Hacker (CEH)
- AWS Educate Machine Learning Foundations - Amazon Web Services
- 5-Star (Golden Badge) in js on Hacker rank
- Google Certified Generative AI and Responsible AI (Google Cloud Badges)
- Meta Back-End Developer Professional Certificate – Coursera